# SAMANALI JAYARATHNA

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## Career Profile

I'm a data scientist with hands-on experience developing end-to-end machine learning pipelines, building AI-driven applications and analysing large-scale datasets. Proficient in Python, SQL, and ML frameworks including scikit-learn and FAISS, with experience in generative AI, RAG systems and LLM integration. Strong foundation in responsible AI practices, cloud technologies and software engineering best practices including GitHub. Experienced in communicating analytical findings clearly to both technical and non-technical stakeholders.

## Education

**Bachelor of Computer Science (Honours),** RMIT University. **July 2026 - July 2027**

**Bachelor of Data Science,** RMIT University. **July 2023 - July 2026**

- GPA: 3.5/4.0 (Pass with Distinction)
- Executive Committee member of RMIT Sri Lanka Association
- General Committee in RMIT SWITCH (Society for Women in Information Technology)

**Foundation in Engineering,** Swinburne University of Technology **July 2022 - June 2023**

- Completed a Foundation Year in Engineering

**Certifications -**

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| RMIT Plus Program- RMIT University | Feb 2026 |
| Programming Foundations: Databases, LinkedIn Learning | Jul 2024 |
| edX (AWS-DA-02), Getting started with Data Analytics on AWS | Dec 2024 |
| edX, Data Science: R Basics | Nov 2023 |

## University Projects

**Eunoia: RAG-Powered Mental Health Chatbot | GitHub Repo: https://github.com/Samanali-J/Eunoia |RMIT**

- Designed and implemented an evaluation framework to measure retrieval quality and response reliability, using prompt grounding as a guardrail to constrain model outputs and reduce hallucination risk.
- Engineered a dual-index retrieval architecture searching a knowledge collection and Q&A bank in parallel, improving context relevance for downstream response generation
- Implemented a prompt-grounding guardrail layer to constrain model outputs to retrieved context, reducing hallucination risk and ensuring reliable, safe responses
- Delivered via an iterative agile workflow using Jira for sprint planning and backlog management, contributing to daily standups and managed version control through feature branches and pull requests on GitHub.

**Key Technologies** - Python, RAG, FAISS, Ollama, LLMs, Flask, sentence embeddings

**Donation Site Operational Hours Monitoring - Power BI Dashboard & Python Preprocessing Pipeline | RMIT WIL**

- Engineered a Python pipeline to extract, clean, and reconcile alarm schedules against donor centre operating hours across 82 fixed Australian Red Cross Lifeblood sites, identifying scheduling discrepancies at scale
- Leveraged the Claude Vision API to automate extraction of structured schedule data from complex multi-page PDFs, replacing a manual validation process that was not feasible across hundreds of pages
- Built a reconciliation engine that compares expected alarm activation times against actual site hours using configurable business rules, flagging mismatches, missing schedules, and internal scheduling conflicts across all site-day combinations
- Delivered a Power BI dashboard with KPI cards, state and day-level breakdowns, and a site deep dive page, enabling the operations team to identify and resolve alarm scheduling errors across the national network

**Key Technologies -** Python, Pandas, Claude Vision API (Anthropic), PyMuPDF, pdfplumber, Power BI, Power Query

**Visual Entailment Classification: Multimodal Deep Learning System | RMIT**

- Built and iteratively evaluated three multimodal architectures (ResNet50 + BERT fusion models) for a visual entailment task, progressing from concatenation-based fusion to cross-attention and self-attention mechanisms
- Applied stratified image-level train, validation, and test splitting to prevent data leakage across correlated samples, and validated model selection against accuracy, AUC, precision, and recall
- Conducted a documented hyperparameter search across learning rate, dropout, and layer-freezing strategies, improving validation AUC from 0.71 to 0.82 through iterative fine-tuning
- Achieved 72% test accuracy and 0.80 AUC on held-out data, demonstrating generalisation beyond the training set

**Key Technologies:** Python, TensorFlow, Keras, BERT, ResNet50, scikit-learn

## Work Experiences

**Content Creator (Part-time), ICS Migration and Education** **Aug 2026 - Present**

- Plan, film and edit short-form video content for the businesss social media pages, taking each piece from concept through to final cut on a regular posting schedule, developing **self-managemen**t by owning the full production process independently and delivering consistently to deadlines without close supervision.
- Act as the on-camera face for a new product launch, presenting to camera and adapting delivery based on feedback from the team and client response, building **resilience and receptiveness to feedback** by learning to separate personal attachment from the work and improve quickly based on what the audience actually responds to.
- Translate complex migration and education service information into accessible, engaging content for a broad audience, strengthening **communication clarity** by stripping back technical or legal language into something a viewer with no prior knowledge can immediately understand and act on.

**Trainee - SAW Engineering Pvt Ltd** **Nov 2025 - Feb 2026**

- Contributed to the environmental sustainability division across solar panel and food waste composting projects, assisting with project coordination and gaining hands-on exposure to clean energy implementation across Sri Lanka, developing **initiative** by proactively seeking involvement across multiple workstreams rather than waiting to be directed.
- Conducted data analysis to support operational and business development decisions, synthesising product specifications and cost data into structured reports for internal stakeholders, building **critical thinking** by learning to identify which information actually mattered for a decision and present it in a way that was useful rather than just comprehensive.
- Represented the company at industry exhibitions across Sri Lanka, engaging directly with clients and industry professionals to promote the product range and build professional relationships, developing **professional confidence** by learning to hold credible conversations with experienced industry figures in an unfamiliar environment far from a classroom setting.

**Retail Assistant - Australian Red Cross, Richmond Bridge Rd** **February 2025 – March 2026**

- Served as first point of contact for a high-traffic charity op shop, managing in-person, phone and email enquiries and resolving most at first contact, building **problem-solving skills** by learning to find workable answers quickly without always having a manager available.
- Guided customers of diverse cultural and linguistic backgrounds through stock, pricing and donation queries, developing **cultural awareness and inclusion** by adapting communication styles to make every customer feel genuinely welcomed.
- Supported a rotating team of volunteers with varying levels of retail experience, sharing register and merchandising knowledge, which developed **informal leadership** by taking responsibility for helping others succeed without holding a formal leadership title.

## Cocurricular activities

**Event Coordinator – RMIT SWITCH** **March 2026 - Present**

- Organised end-to-end logistics for tech-focused events including networking sessions, coding competitions, and industry collaborations with partners such as Notion.

**Peer Mentor – Machine Learning, RMIT University** **Feb 2026 - Present**

- Mentored students on model evaluation, overfitting, and implementation trade-offs, bridging theoretical concepts with practical Python application.

**Publicity Officer - RMIT Sri Lankan Association** **August 2024 – August 2026**

- Managed the association's social media presence and led publicity for large-scale cultural events attracting hundreds of attendees

## Skills

**Technical:**

- **Programming Languages:** Python, R, SQL
- **Data Tools:** Pandas, NumPy, Matplotlib, seaborn, Plotly, scikit-learn, , TensorFlow, Keras, Excel, PowerBI
- **Databases:** MySQL, MongoDB
- **Cloud & Tools:** AWS (EC2, S3, Lambda, ECS),Docker, GitHub, Google Colab, Flask
- **Project & Collaboration Tools:** Jira, Trello, draw.io, Microsoft Office Suite
- **Engineering Practices:** ETL/ELT pipeline development, Agile delivery (sprint planning, Kanban), version control(GitHub)
- **Security & Forensics Tools:** Autopsy,FTK, CISCO

**Other**: Microsoft Office Suite

**Languages:** English, Sinhala

## Referees

Referees available upon request